



WeCloudData

Machine Learning Workflow



Agenda

Machine Learning Workflow

3 ML Workflow

16 ML Workflow with code demo

30 ML Workflow In-depth Example



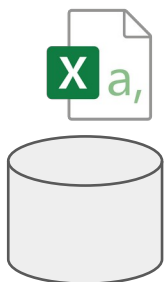


ML Workflow



Machine Learning Workflow

Understanding the ML Workflow





1. Loading Data (INPUT)

Understanding the ML Workflow





2. Label Data

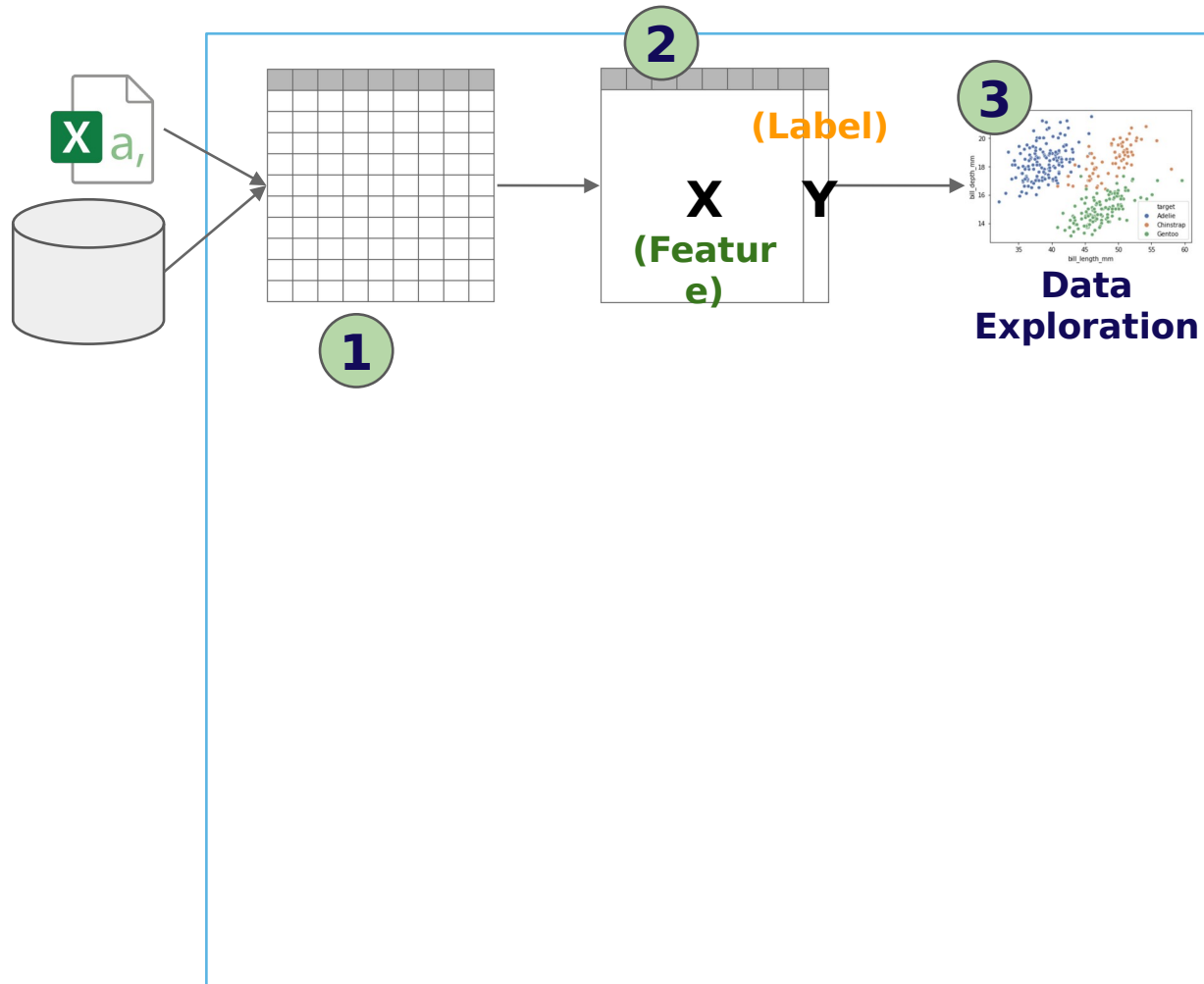
Understanding the ML Workflow





3. Explore Data

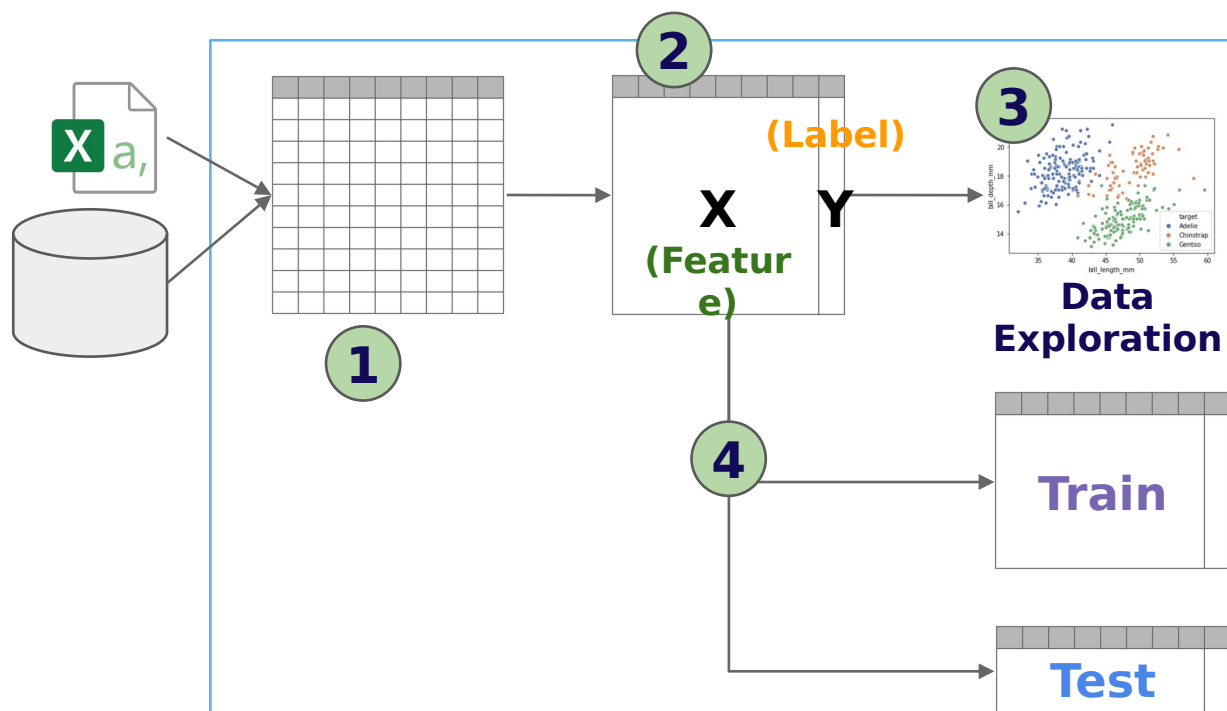
Understanding the ML Workflow





4. Split Data into Train/Test

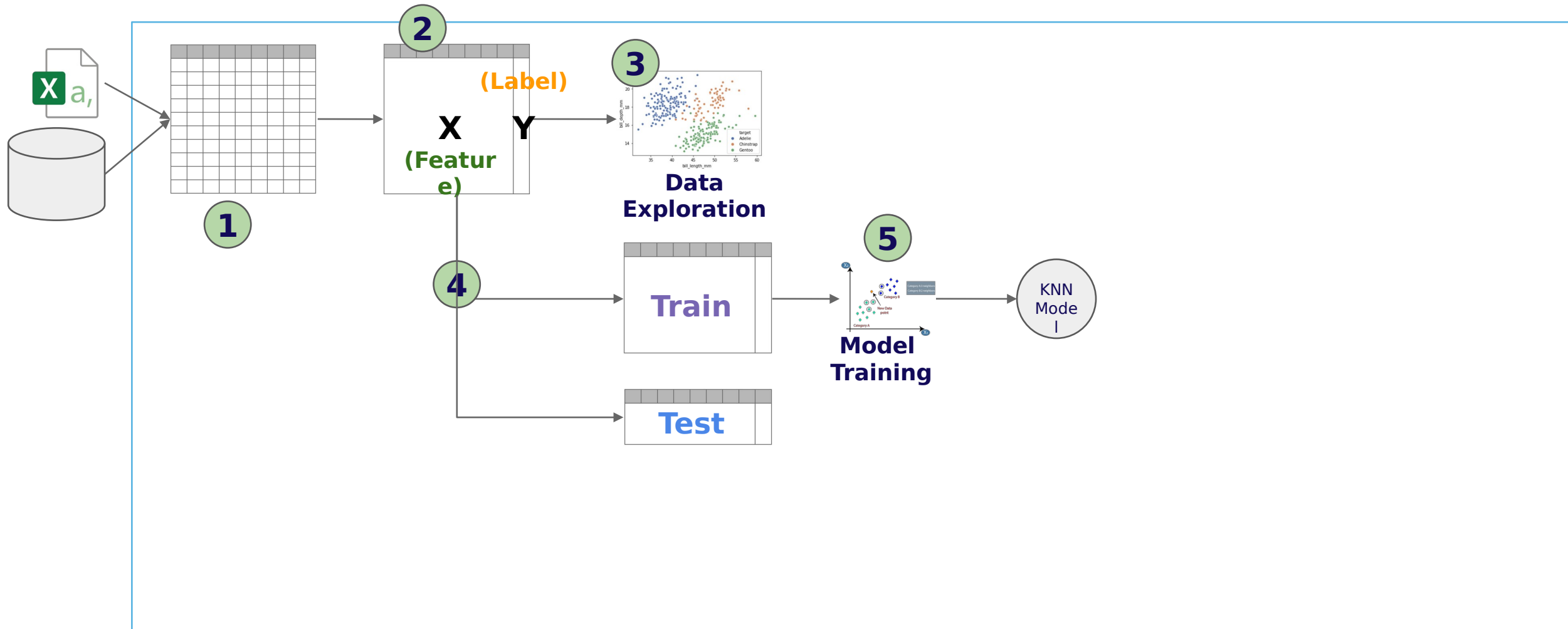
Understanding the ML Workflow





5. Train Models

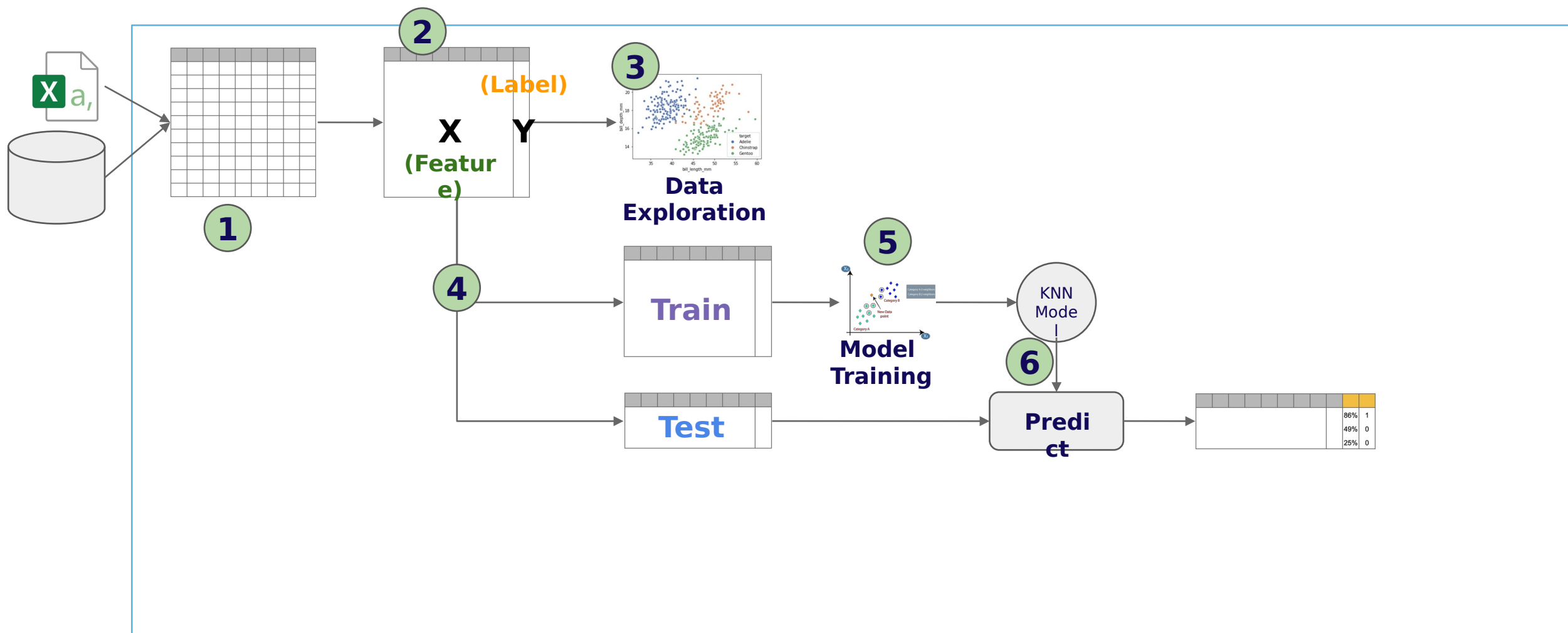
Understanding the ML Workflow





6. Make Prediction on Test Data

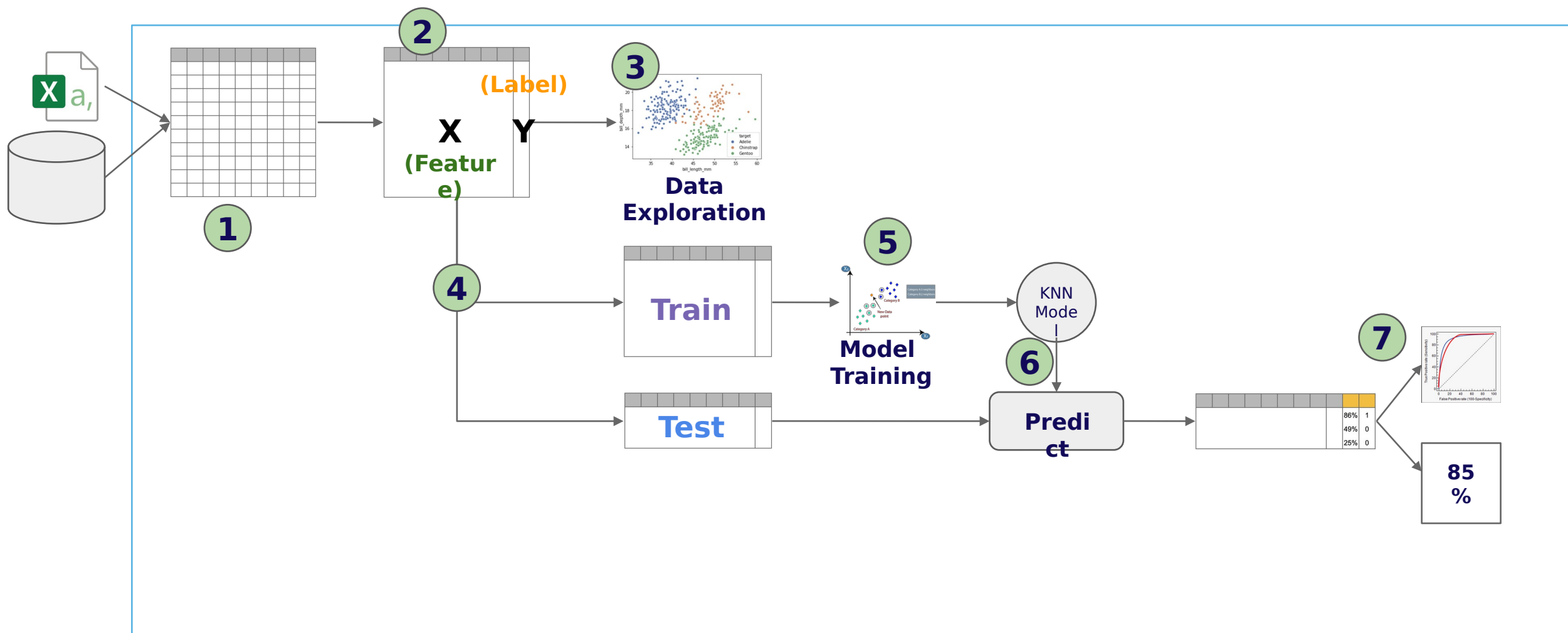
Understanding the ML Workflow





7. Evaluate the Model Prediction

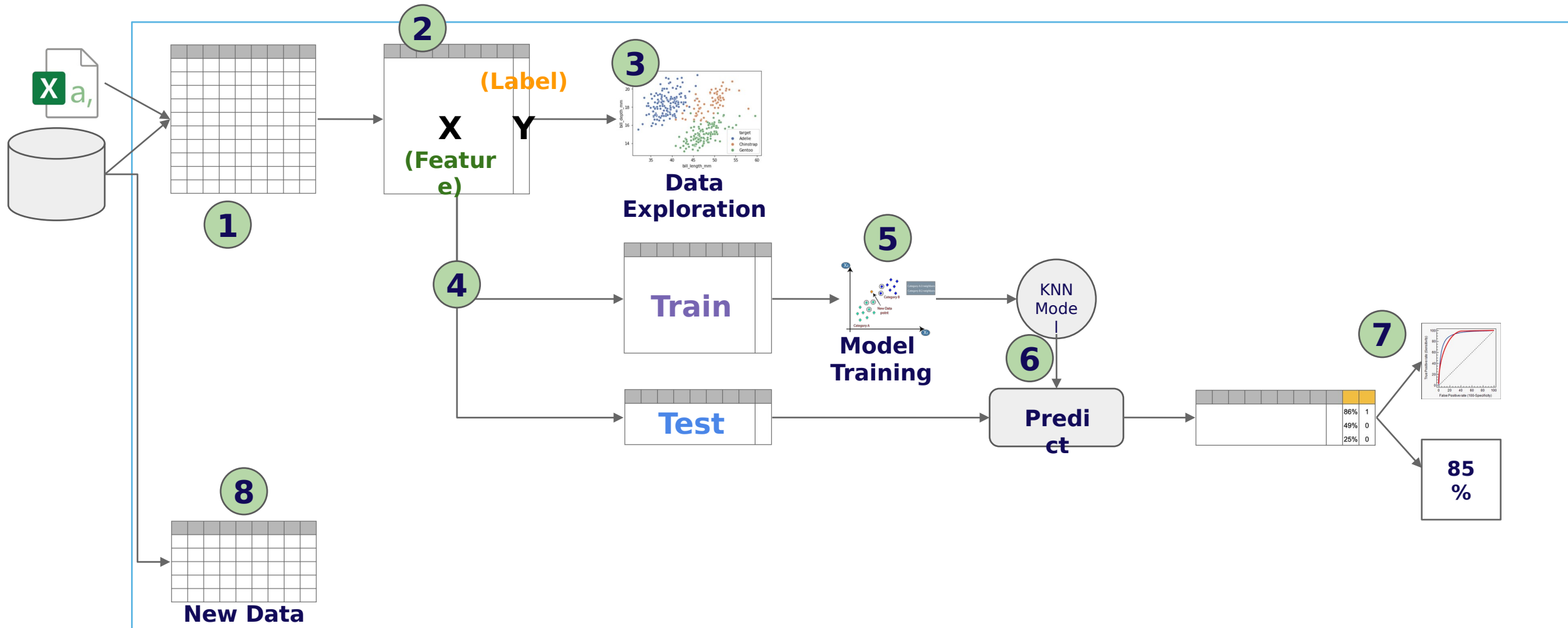
Understanding the ML Workflow





8. Load New/Unseen Data

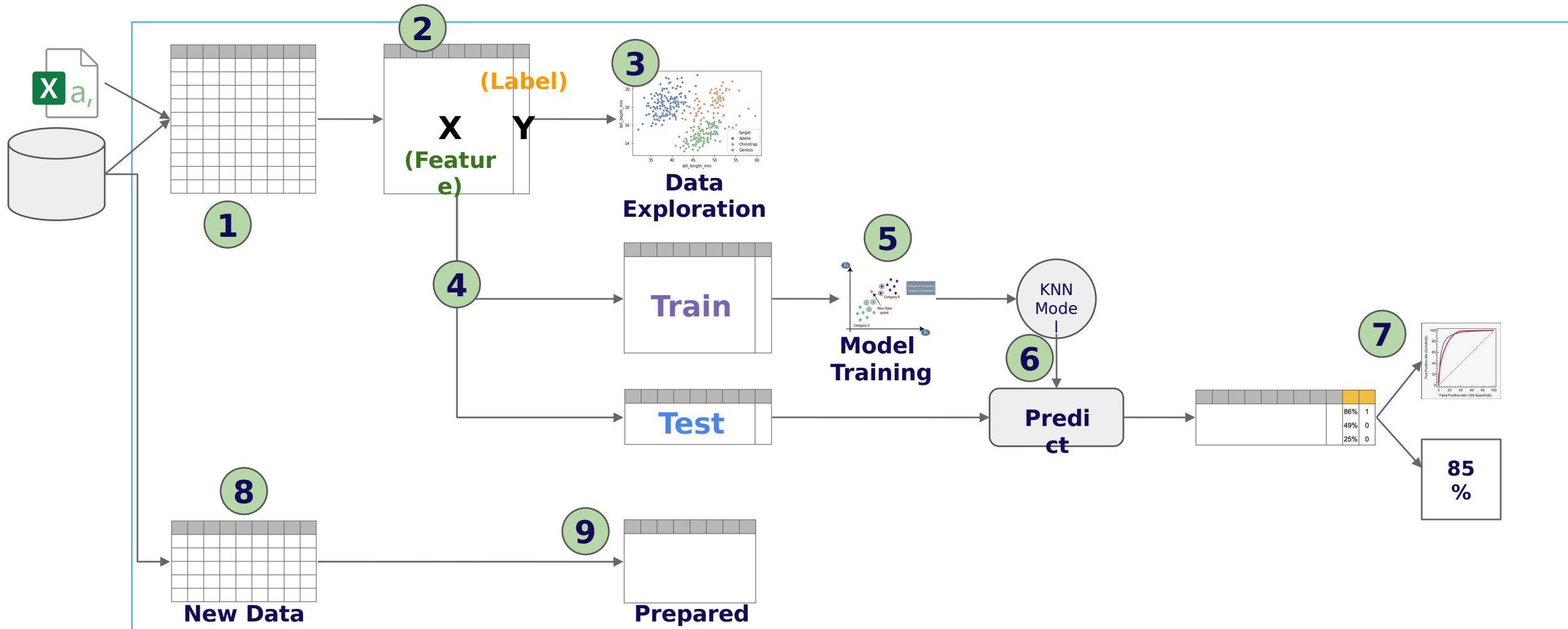
Understanding the ML Workflow





9. Prepare the New Data

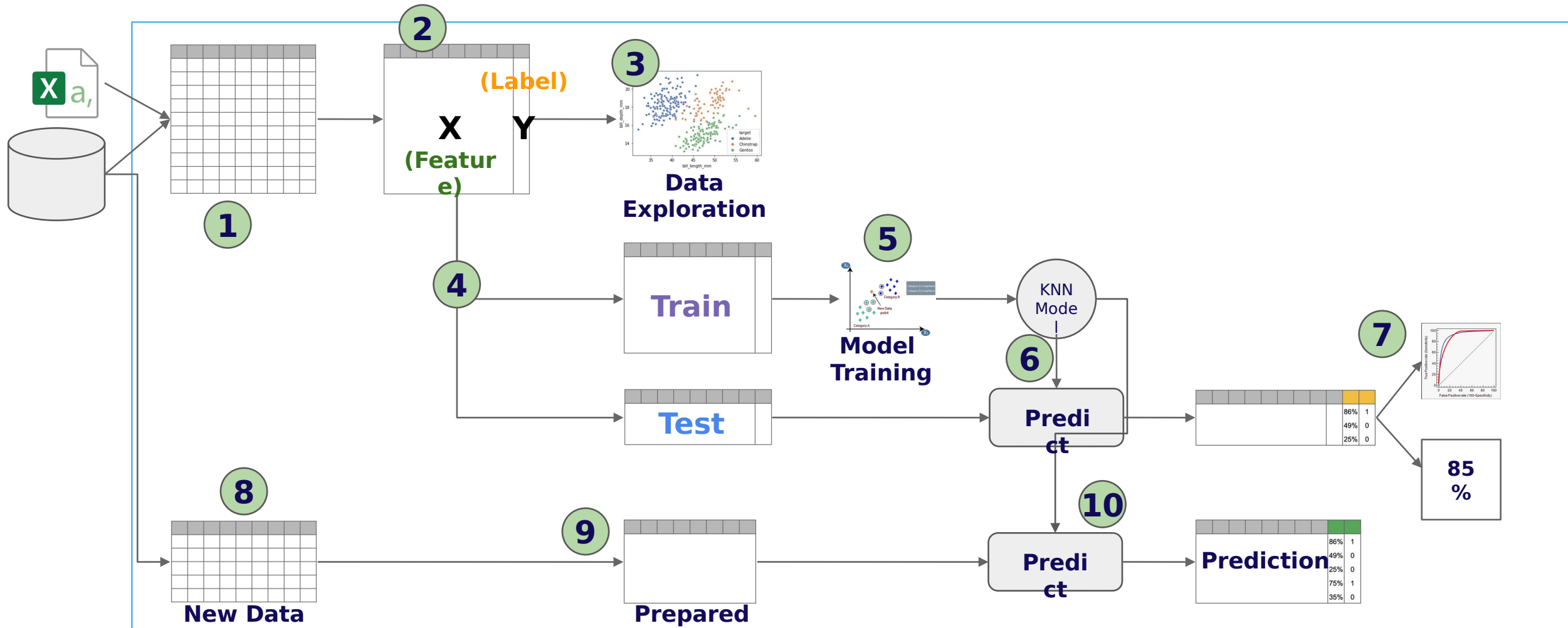
Understanding the ML Workflow





10. Generate Prediction on New Data (OUTPUT)

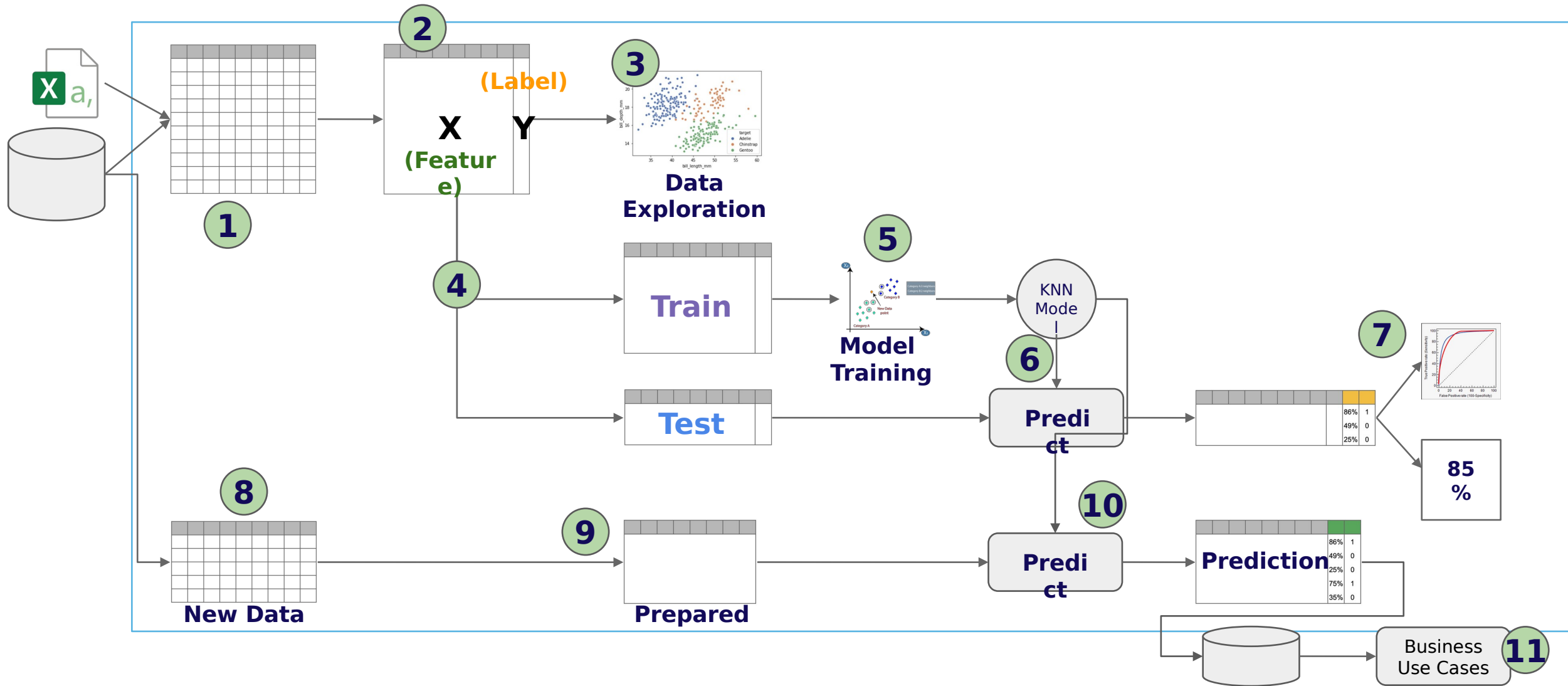
Understanding the ML Workflow





11. Use the Model Prediction

Understanding the ML Workflow





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ML Workflow with code demo

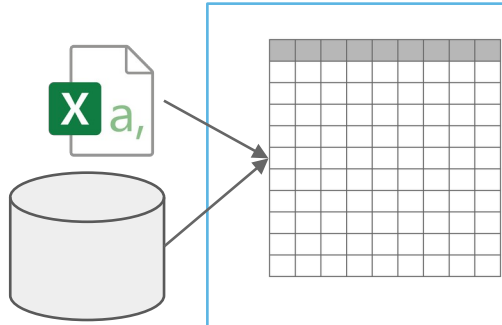


Machine Learning Process - Beginner

Understanding the ML Workflow

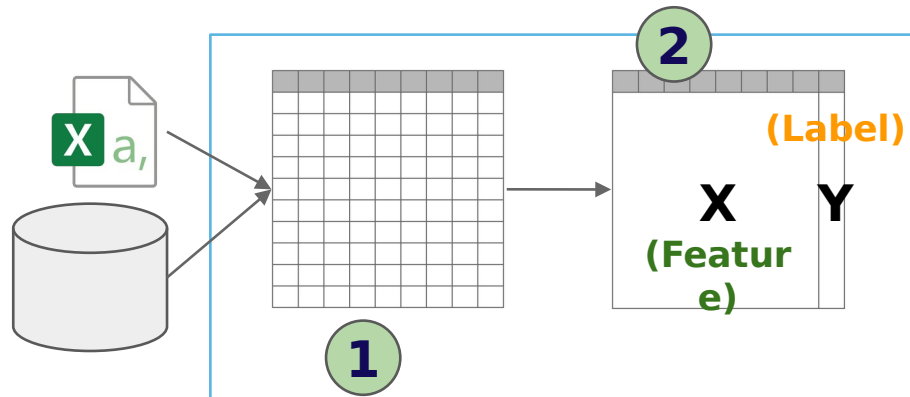
**Let's go through the workflow
again,
with code demo**

1. Load Data



```
# step 1 - load data  
iris = pd.read_csv('iris/iris.csv')
```

2. Label Data

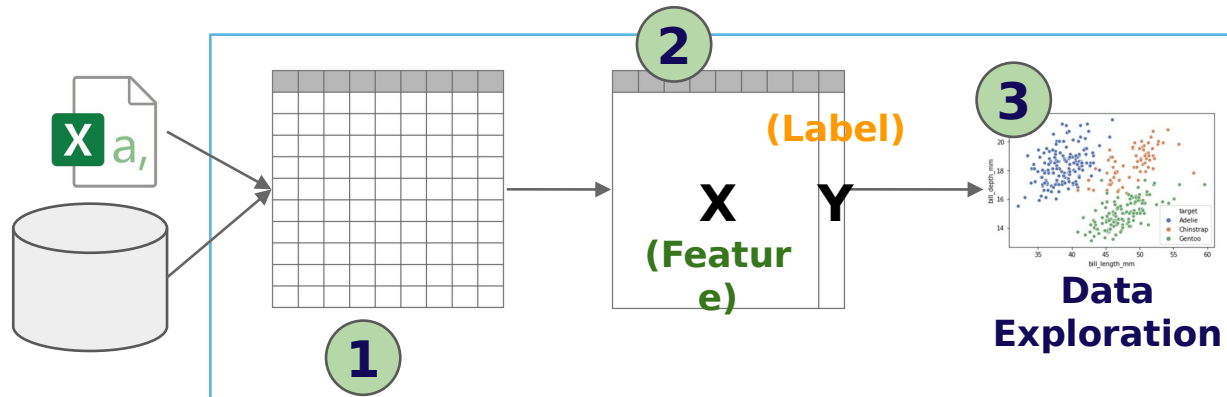


```
# step 2 - label data
```

```
X = iris.loc[:, iris.columns.drop('Species')]
```

```
y = pd.Series(iris.Species).map({'setosa': 0, 'versicolor': 0, 'virginica': 1})  
    .to_frame()  
    .rename(columns={0: 'target'})
```

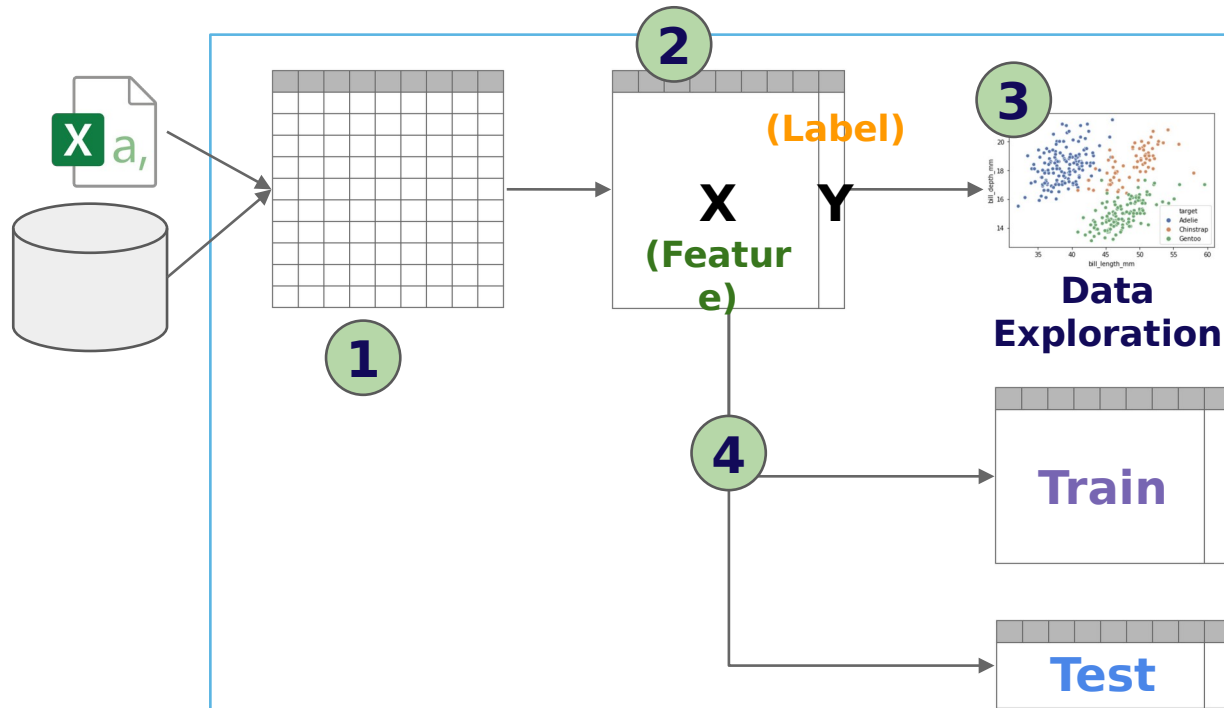
3. Explore Data



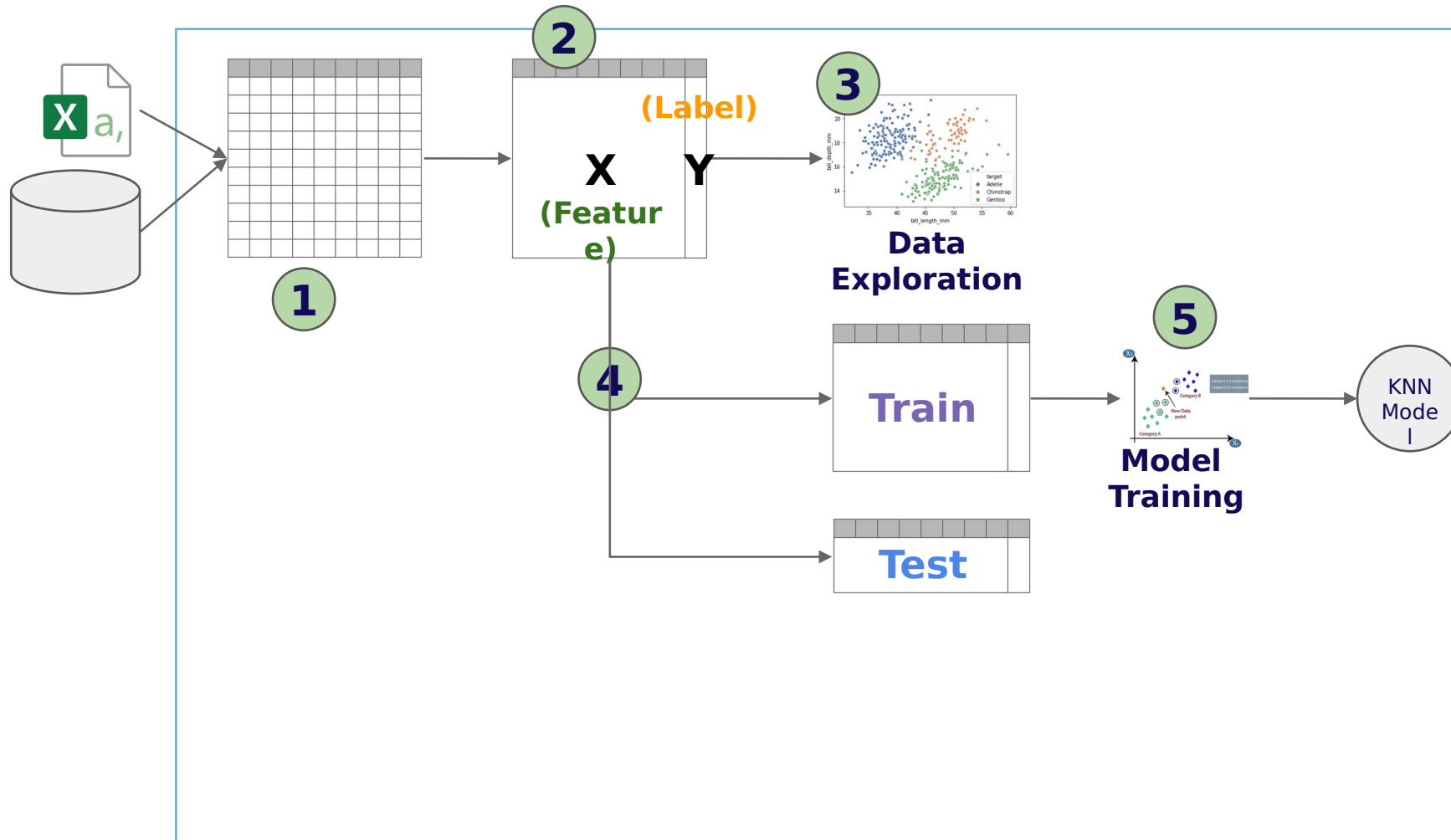
```
# step 3 - explore data
```

```
X.plot.scatter(x="Sepal.Length", y="Sepal.Width", figsize=(6,4))
```

4. Split Data

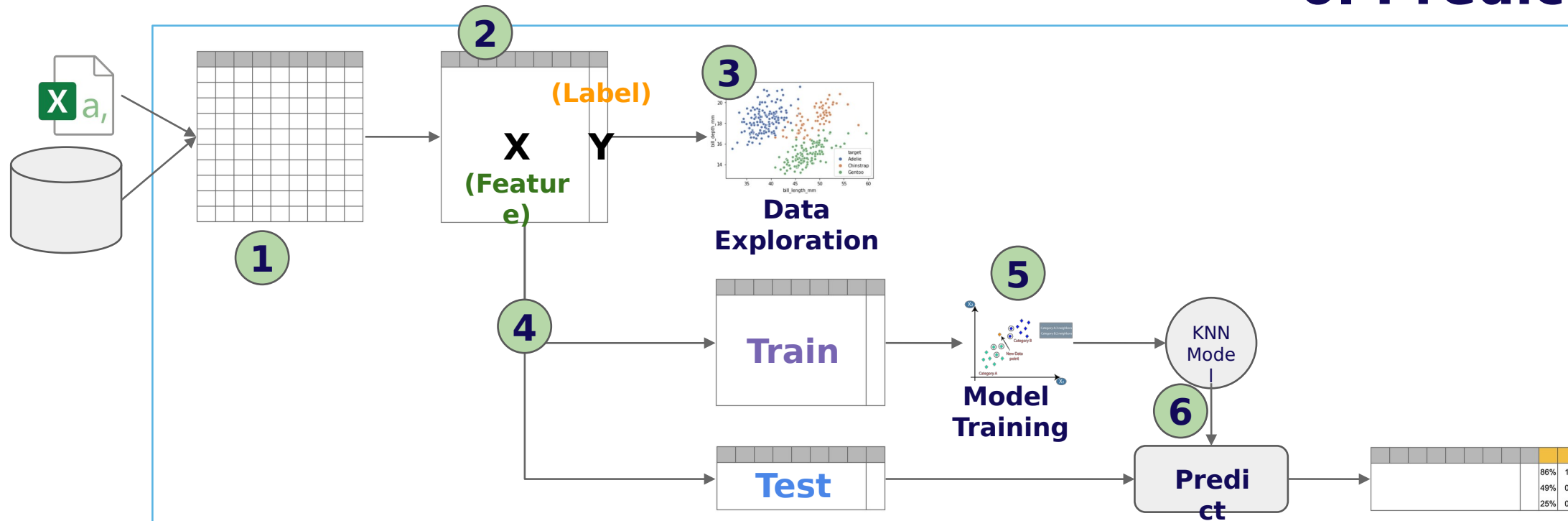
[illegible]

5. Model Training



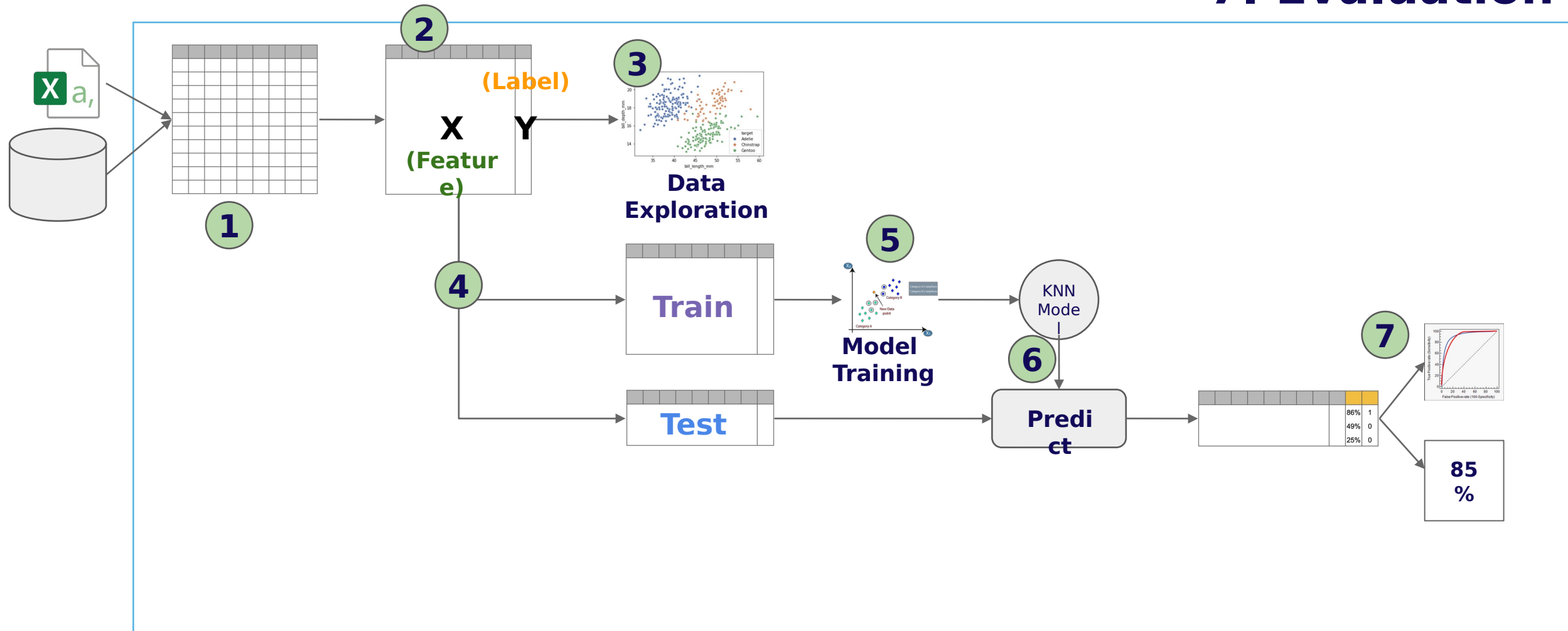
```
# step 5 - model training
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors=10)
knn.fit(X_train, y_train)
```

6. Prediction on Test



```
# step 6 - generate prediction
predictions = knn.predict(X_test)
```

7. Evaluation

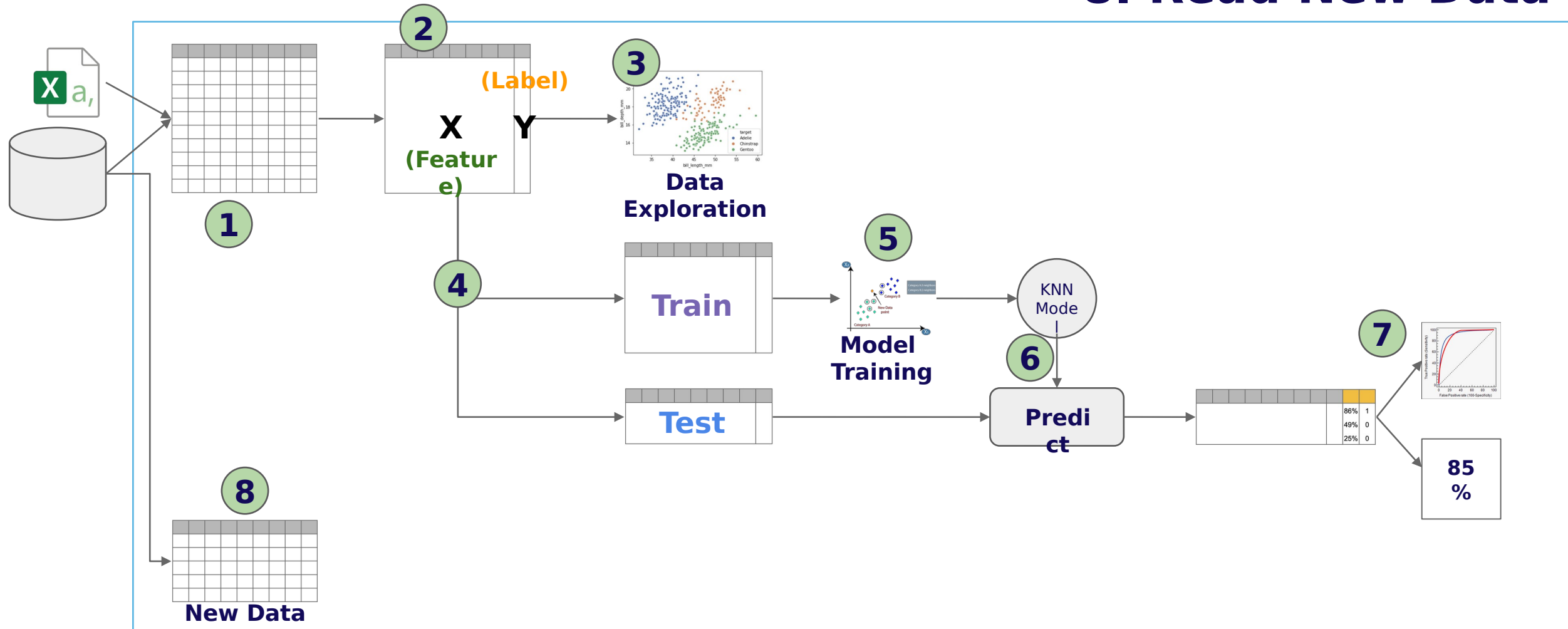


```
# step 7 - model evaluation
```

```
accuracy = sum(predictions == y_test.Species.values) / float(len(y_test))
```

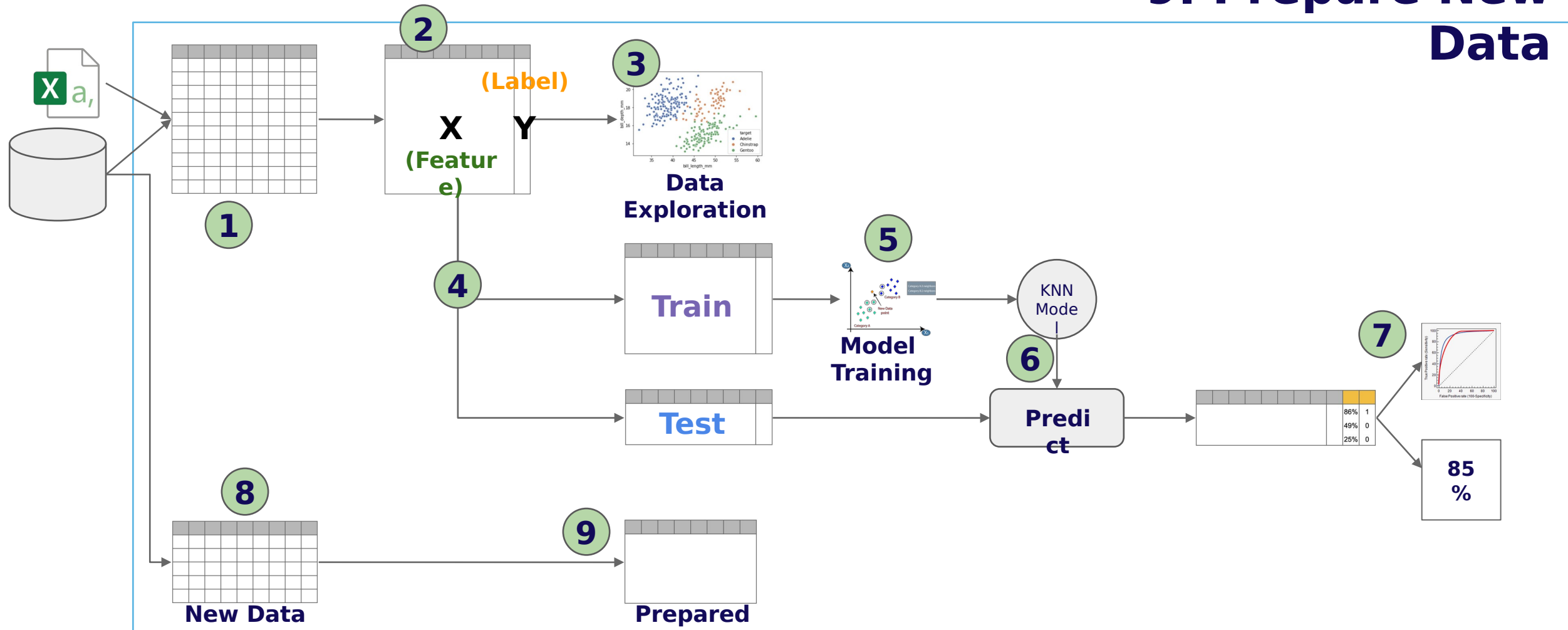
```
print (accuracy)
```


8. Read New Data



```
# step 8 - load new data
iris_new = pd.read_csv('iris_new.csv')
```

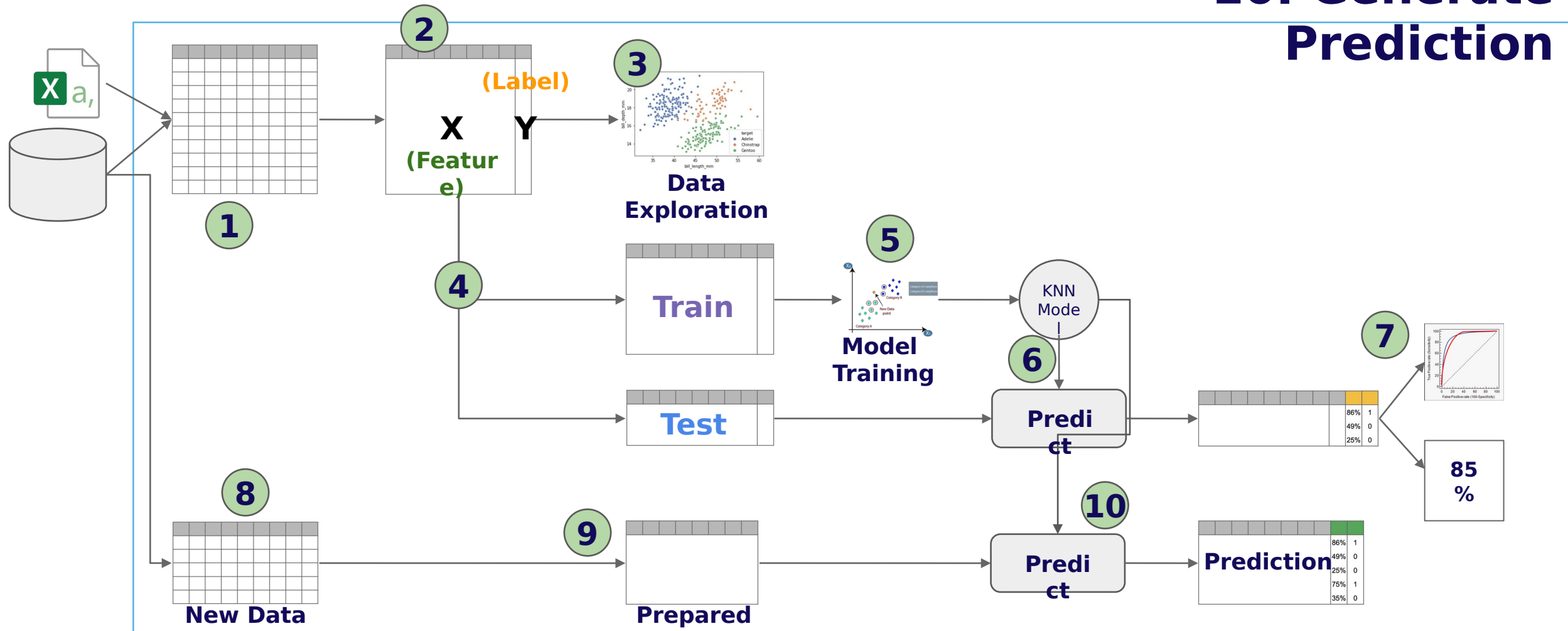
9. Prepare New Data



step 9 - prepare new data

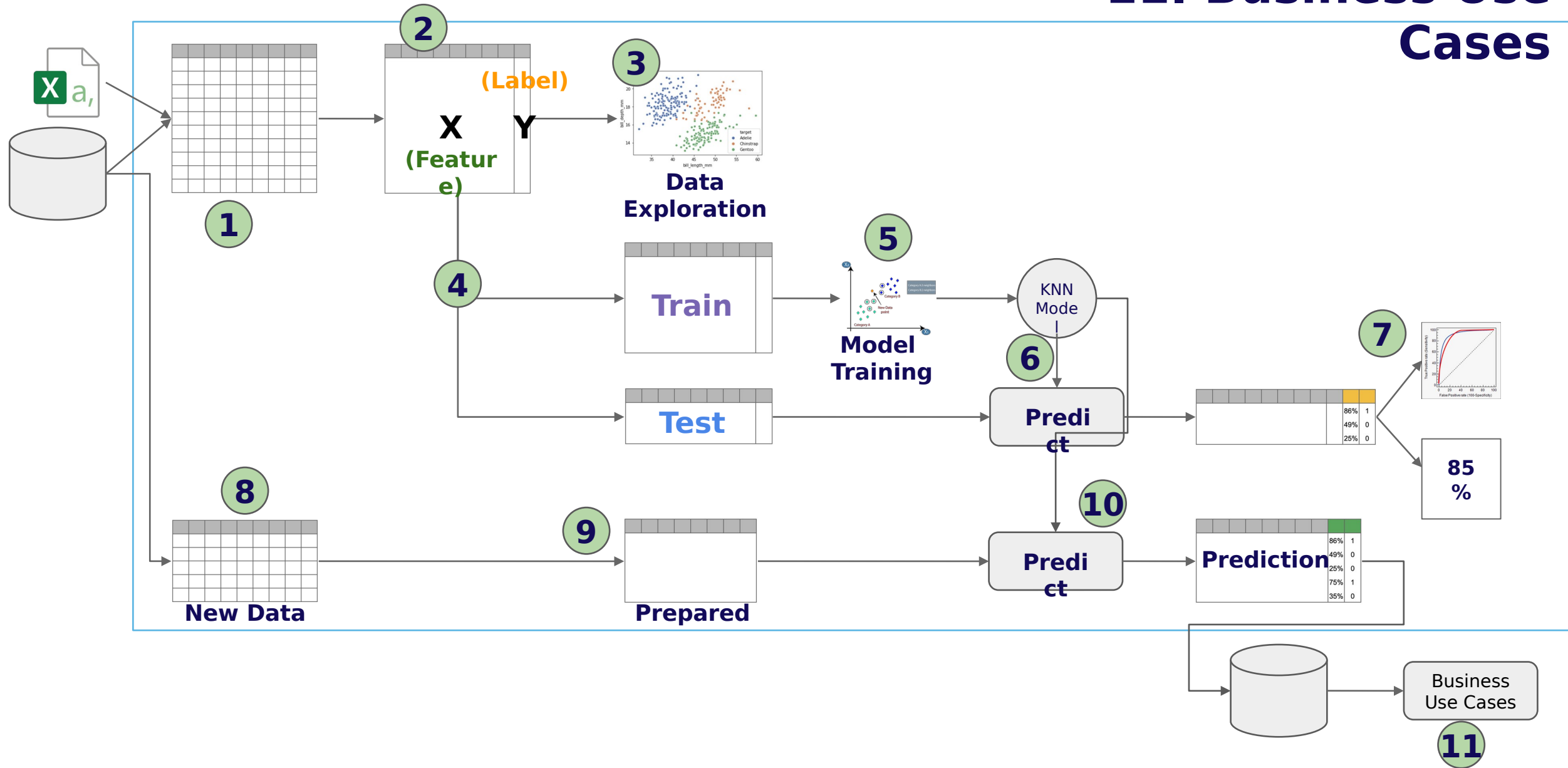
```
new_prepared = iris_new.loc[:, iris_new.columns.drop('Species')]
```

10. Generate Prediction



```
# step 10 - make predictions on new data
new_prediction = knn.predict(new_prepared)
print(new_prediction)
```

11. Business Use Cases





Machine Learning

Understanding the ML Workflow

```
%matplotlib inline
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

# step 1 - load data
iris =
pd.read_csv('https://s3.amazonaws.com/weclouddata/datasets/random/
iris/iris.csv')

# step 2 - label data
X = iris.loc[:, iris.columns.drop('Species')]

y = pd.Series(iris.Species).map({'setosa': 0, 'versicolor': 0,
'virginica': 1}).to_frame().rename(columns={0: 'target'})

# step 3 - explore data
X.plot.scatter(x="Sepal.Length", y="Sepal.Width", figsize=(6,4))

# step 4 - split data into train vs test
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, shuffle=True, random_state=2018)
```

```
# step 5 - model training
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors=10)
knn.fit(X_train, y_train)

# step 6 - generate prediction
predictions = knn.predict(X_test)

# step 7 - model evaluation
accuracy = sum(predictions == y_test.Species.values) /
float(len(y_test))
print (accuracy)

# step 8 - load new data
iris_new =
pd.read_csv('https://s3.amazonaws.com/weclouddata/datasets/random/
iris/iris.csv')

# step 9 - prepare new data
new_prepared = iris_new.loc[:, iris_new.columns.drop('Species')]

# step 10 - make predictions on new data
new_prediction = knn.predict(new_prepared)
print(new_prediction)
```

<https://pinetools.com/syntax-highlighter>



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ML Workflow In-depth Example

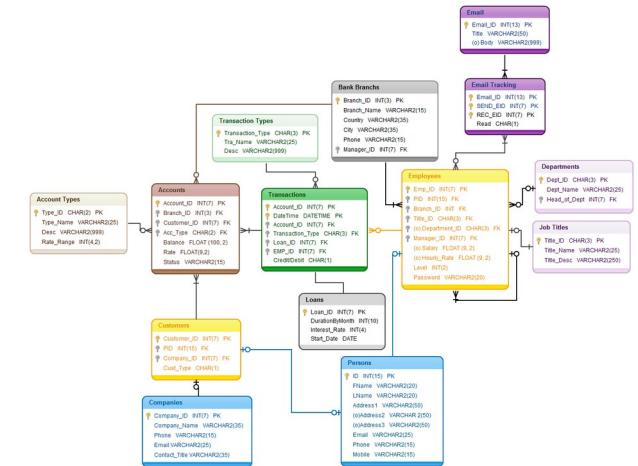


Machine Learning

Understanding the ML Workflow

Too abstract?

Let's look at an example



Data Collection
Data Preparation
Data Exploration
Feature Engineering
Modeling
Validation
Interpretation
Deployment

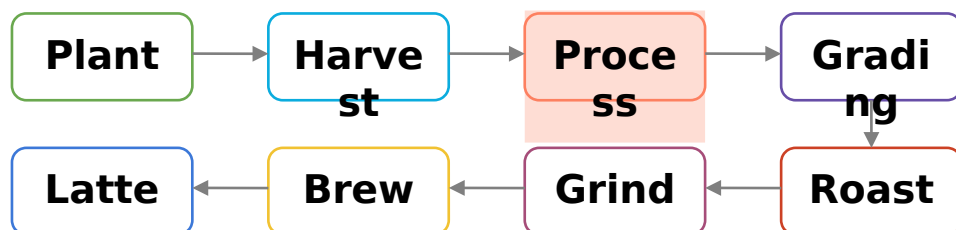




Data Preparation

Understanding the ML Workflow

Data Collection
Data Preparation
Data Exploration
Feature Engineering
Modeling
Validation
Interpretation
Deployment



- Data gets cleaned and transformed into a ML-friendly format
- Common data prep tools include
 - Excel, SQL, Python
 - No-Code data prep tool such as Sagemaker

ID (Index)	Labels (Target)						
	Credit Approval	Age	Gender	Annual Salary	Months in Residence	Months in Job	Current Debt
	Client 1	23	M	\$30,000	36	12	\$5,000
	Client 2	30	F	\$45,000	12	12	\$1,000
	Client 3	19	M	\$15,000	3	1	\$10,000
	Client 4	25	M	\$25,000	12	27	\$15,000
Paid off Credit							Yes
							Yes
							No
							?

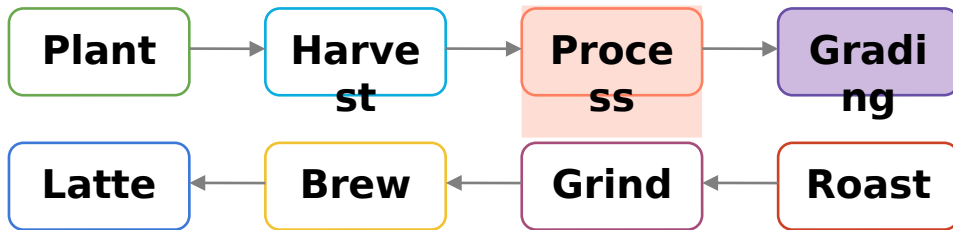
Features (Predictors | Input Variables)



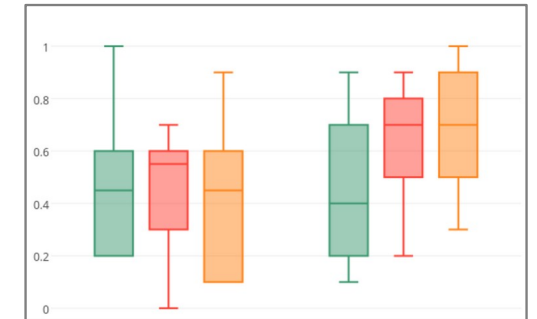
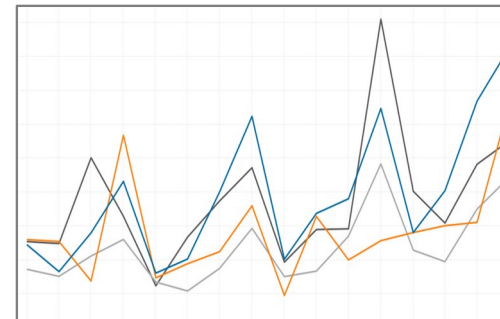
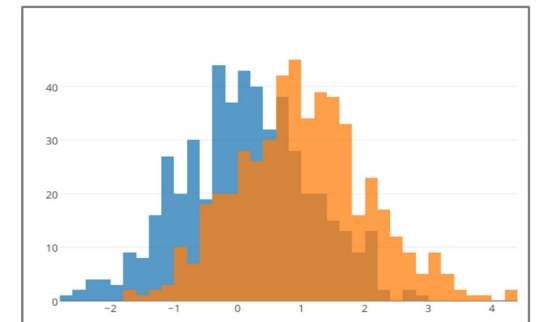
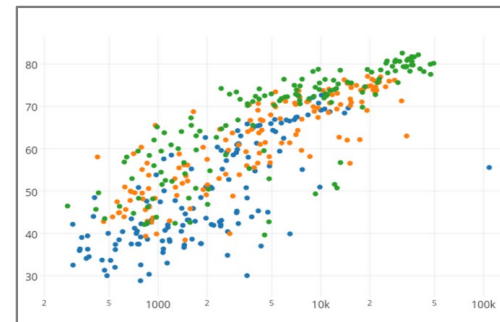
Data Exploration

Understanding the ML Workflow

Data Collection
Data Preparation
Data Exploration
Feature Engineering
Modeling
Validation
Interpretation
Deployment



- Data visualization and exploration is a key step to understand about the data problems
- Tools include: Excel, Tableau, PowerBI, SageMaker, Python

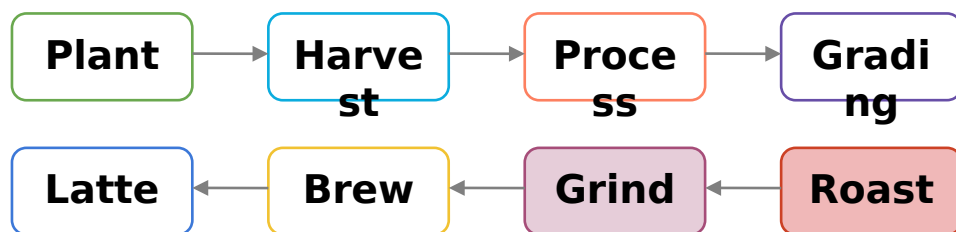




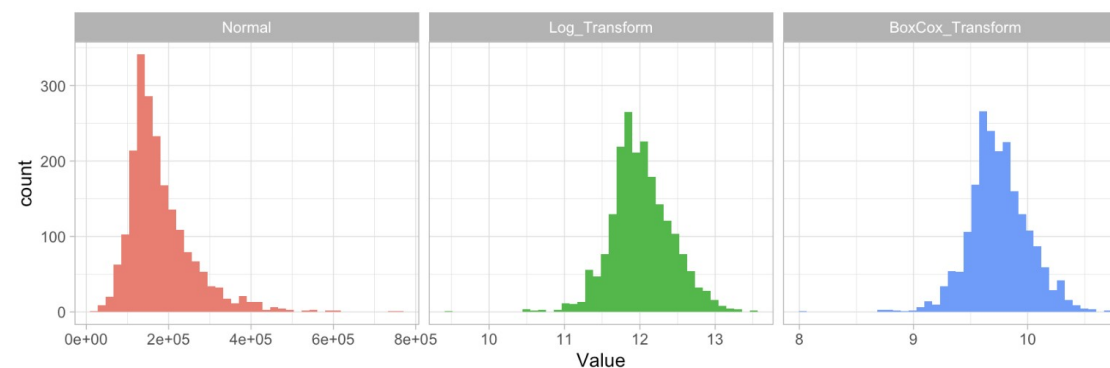
Feature Engineering

Understanding the ML Workflow

Data Collection
Data Preparation
Data Exploration
Feature Engineering
Modeling
Validation
Interpretation
Deployment



- Data gets turned into features which are then engineered to empower high-performance models
- Feature preparation tools include:
 - SQL, Python, R, Sagemaker, etc.



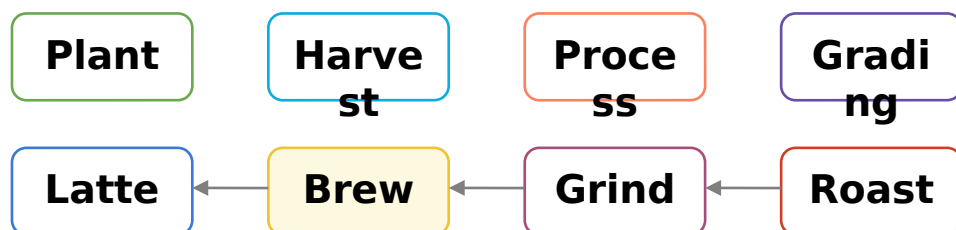
<https://bradleyboehmke.github.io/HOML/engineering.html>



Modeling

Understanding the ML Workflow

Data Collection
Data Preparation
Data Exploration
Feature Engineering
Modeling
Validation
Interpretation
Deployment



- During the modeling stage, different ML algorithms are tried and the best one is selected
- ML tools:
 - Python, PyTorch, Tensorflow, SageMaker, etc.

COFFEE BREWING METHODS



Dorian Alsela
IL BARISTA



Modeling: Linear Regression

Understanding the ML Workflow

Data Collection

Data Preparation

Data Exploration

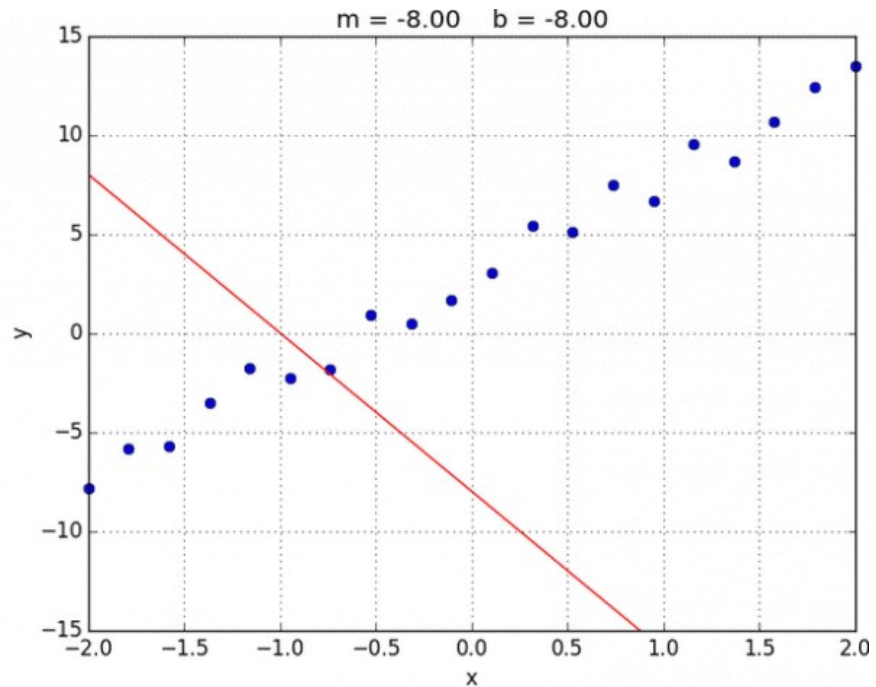
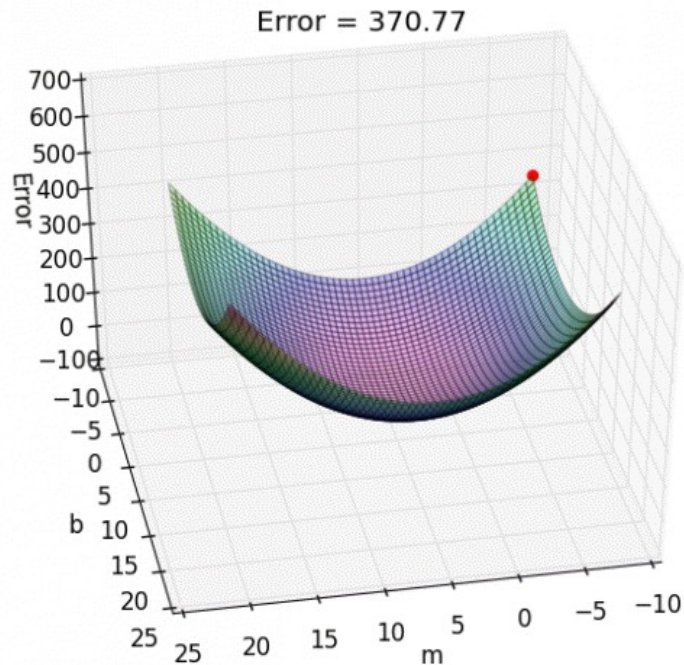
Feature Engineering

Modeling

Validation

Interpretation

Deployment





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Demo: Linear Regression

A Visual Introduction to Linear Regression



```

graph TD
    Root["petal length (cm) <= 2.45  
gini = 0.667  
samples = 150  
value = [50, 50, 50]  
class = setosa"]
    Root --> L1["gini = 0.0  
samples = 50  
value = [50, 0, 0]  
class = setosa"]
    Root --> R1["petal width (cm) <= 1.75  
gini = 0.5  
samples = 100  
value = [0, 50, 50]  
class = versicolor"]
    R1 --> L2["petal length (cm) <= 4.95  
gini = 0.168  
samples = 54  
value = [0, 49, 5]  
class = versicolor"]
    R1 --> R2["petal length (cm) <= 4.85  
gini = 0.043  
samples = 46  
value = [0, 1, 45]  
class = virginica"]
    L2 --> L3["petal width (cm) <= 1.65  
gini = 0.041  
samples = 48  
value = [0, 47, 1]  
class = versicolor"]
    L2 --> L4["petal width (cm) <= 1.55  
gini = 0.444  
samples = 6  
value = [0, 2, 4]  
class = virginica"]
    L3 --> L5["gini = 0.0  
samples = 47  
value = [0, 47, 0]  
class = versicolor"]
    L3 --> L6["gini = 0.0  
samples = 1  
value = [0, 0, 1]  
class = virginica"]
    L4 --> L7["gini = 0.0  
samples = 3  
value = [0, 0, 3]  
class = virginica"]
    L4 --> L8["sepal length (cm) <= 6.95  
gini = 0.444  
samples = 3  
value = [0, 2, 1]  
class = versicolor"]
    L8 --> L9["gini = 0.0  
samples = 2  
value = [0, 2, 0]  
class = versicolor"]
    L8 --> L10["gini = 0.0  
samples = 1  
value = [0, 0, 1]  
class = virginica"]
    R2 --> L11["sepal width (cm) <= 3.1  
gini = 0.444  
samples = 3  
value = [0, 1, 2]  
class = virginica"]
    R2 --> L12["gini = 0.0  
samples = 43  
value = [0, 0, 43]  
class = virginica"]
    L11 --> L13["gini = 0.0  
samples = 2  
value = [0, 0, 2]  
class = virginica"]
    L11 --> L14["gini = 0.0  
samples = 1  
value = [0, 1, 0]  
class = versicolor"]
  
```



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Demo: Decision Tree

[A visual explanation of decision tree](#)



Modeling: Neural Networks

Understanding the ML Workflow

Data Collection

Data Preparation

Data Exploration

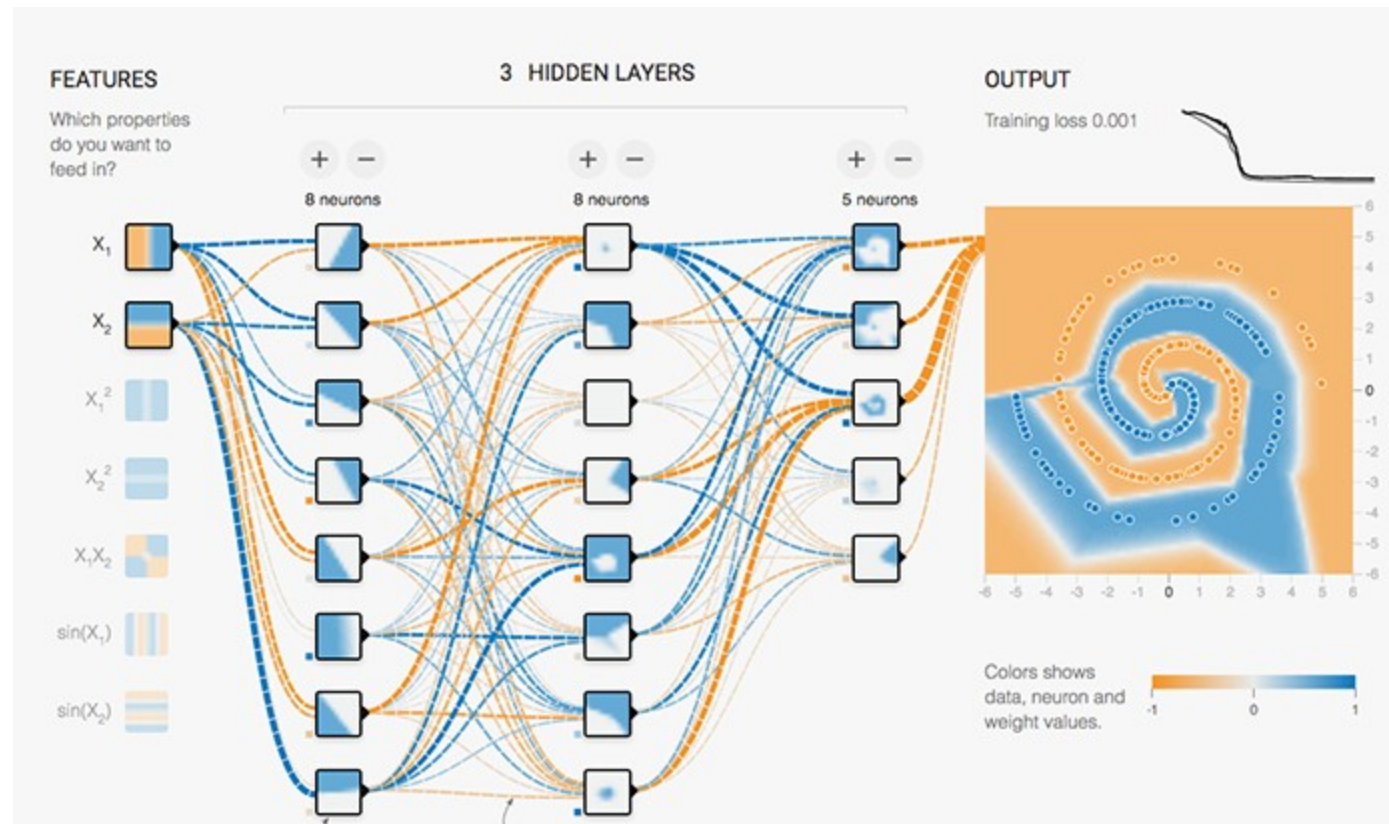
Feature Engineering

Modeling

Validation

Interpretation

Deployment





WeCloudData

Demo: Neural Networks

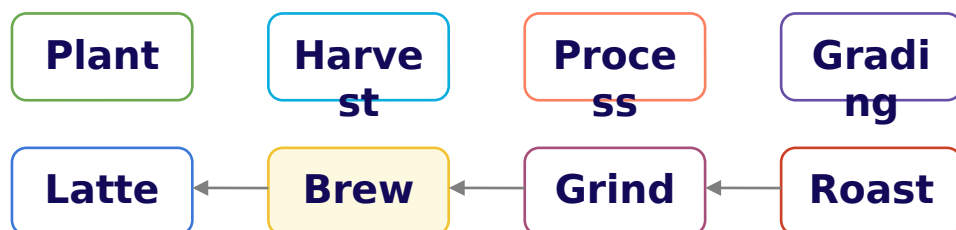
[Neural network playground](#)



Evaluation & Validation

Understanding the ML Workflow

Data Collection
Data Preparation
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Modeling
Validation
Interpretation
Deployment



How do we know that the model works?

Model Evaluation

- We need a set of metrics to benchmark the model performance

Model Validation

- We also want to know that the model is well tested and the good result is reproducible.



Model Evaluation

Understanding the ML Workflow

Data Collection

Data Preparation

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Feature Engineering

Modeling

Validation

Interpretation

Deployment

Confusion Matrix

Good Model

Actual	Positive	Negative
	23	7
Predicted	Positive	10
	Negative	60

Faulty Model

Actual	Positive	Negative
	23	7
Predicted	Positive	58
	Negative	12



Model Evaluation

Understanding the ML Workflow

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Precision & Recall

Precision

		Real Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Zeyu, 2021

Recall

		Real Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

Zeyu, 2021

Source: Precision and recall made simple ([link](#))



Model Evaluation

Understanding the ML Workflow

Data Collection

Data Preparation

Data Exploration

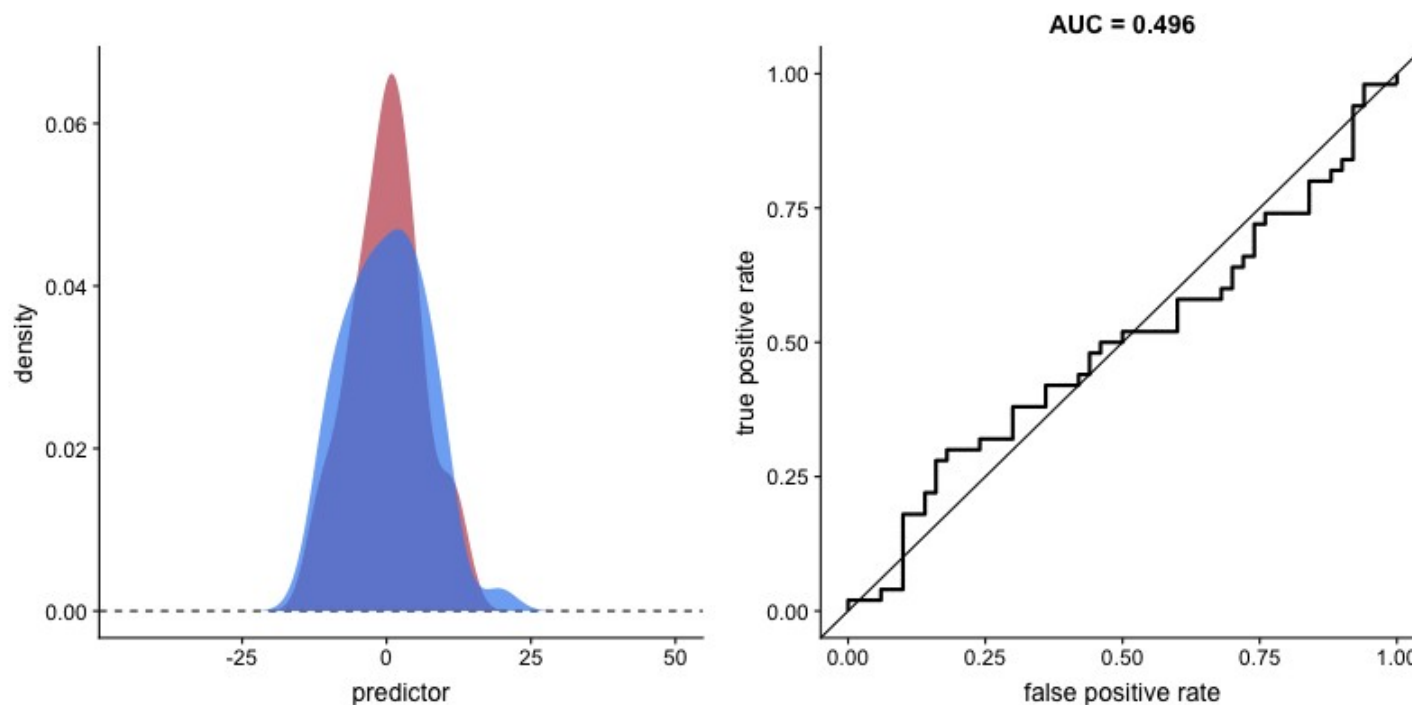
Feature Engineering

Modeling

Validation

Interpretation

Deployment



ROC

Source: ROC, AUC, precision, and recall visually explained ([link](#))

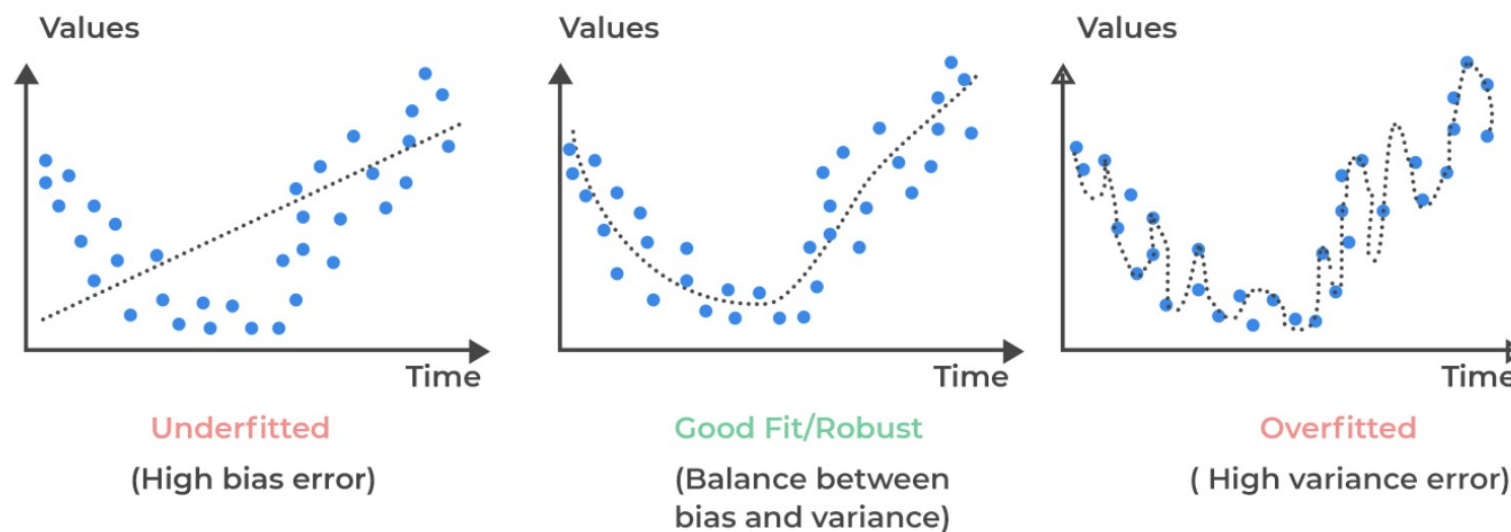


Model Validation

Understanding the ML Workflow

Data Collection
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Validation
Interpretation
Deployment

Generalization & Fit



Source: Overfitting and methods of addressing it ([link](#))

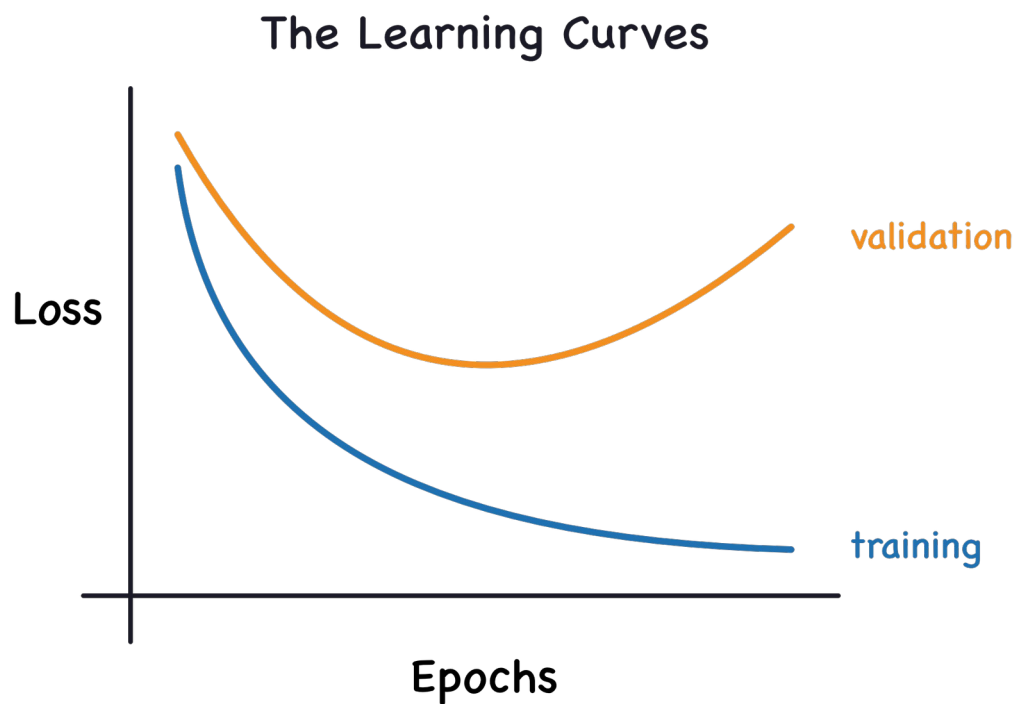


Model Validation

Understanding the ML Workflow

Data Collection
Data Preparation
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Validation
Interpretation
Deployment

Learning Curves



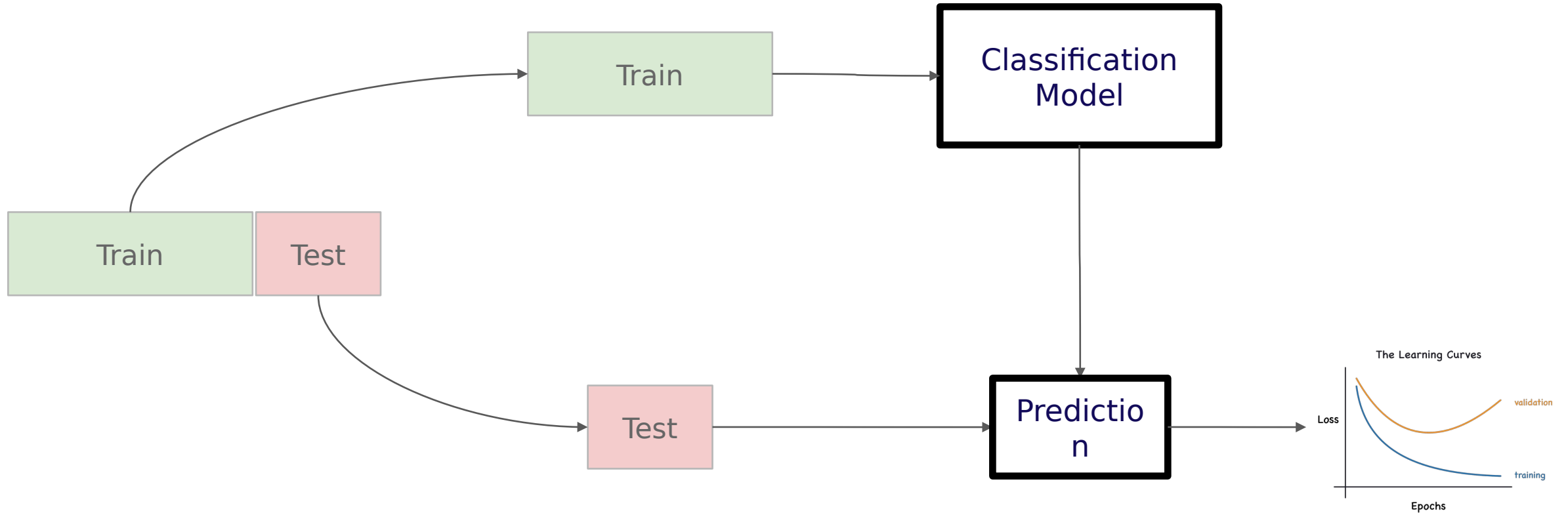


Model Validation

Understanding the ML Workflow

Data Collection
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Validation
Interpretation
Deployment

Data Splitting





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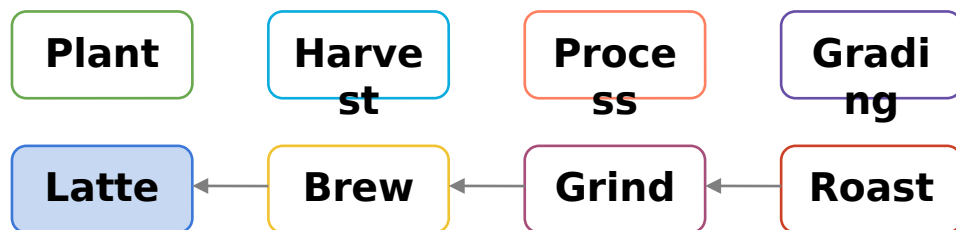
Demo: Evaluation & <https://mlu-explain.github.io/train-test-validation> Validation



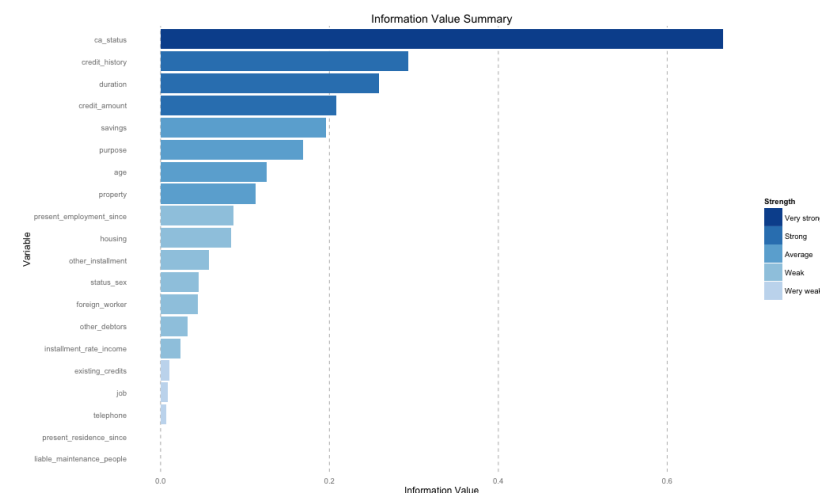
Interpretation

Understanding the ML Workflow

Data Collection
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How do we use the model? What features influence the model prediction?

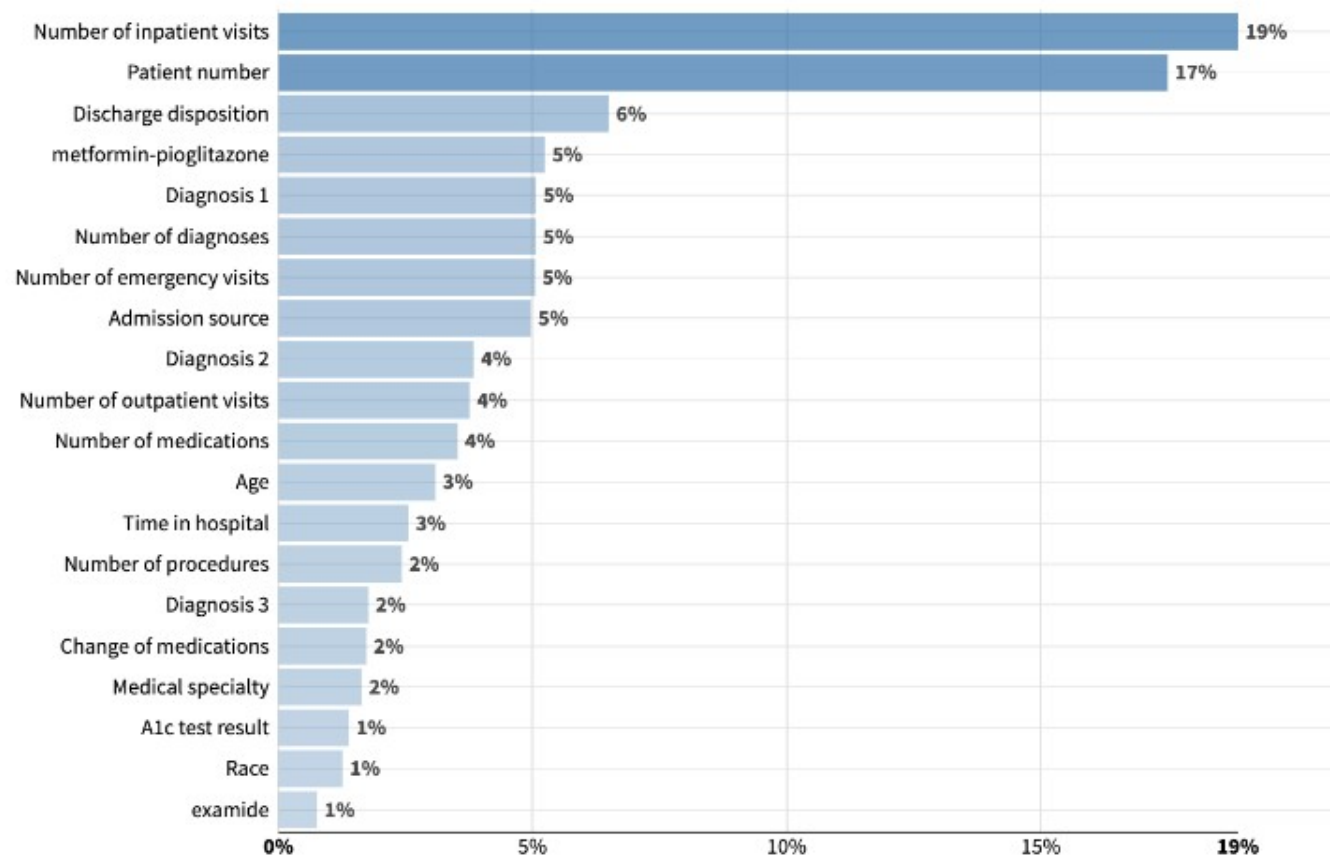




Interpretation - Feature Importance

Understanding the ML Workflow

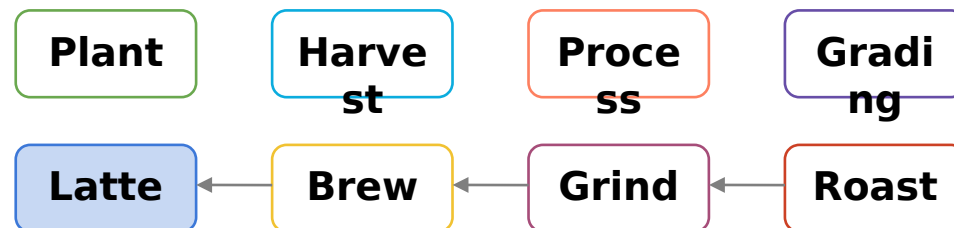
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Deployment

Understanding the ML Workflow



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!